In [4]:
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# Hemant Marotkar ,UID:- 23015002
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

#Load Dataset
df = pd.read_csv("basic_cyber_dataset.csv")

#Display first 5 rows
print("First 5 rows:")
print(df.head())

#Dataset shape
print("\nDatasetsize:")
print(df.shape)

#Column names
print("\nColumn names:")
print(df.columns)

#Dataset information
print("\nDataset information:")
print(df.info())

#Staistical summary
print("\nStatistical summary:")
print(df.describe())

#Missing values
print("\nMissing values:")
print(df.isnull().sum())

#Class distribution
print("\nClass distribution:")
print(df["label"].value_counts())

#Visualization
df["label"].value_counts().plot(kind="bar")
plt.title("Normal vs Attack Traffic")
plt.xlabel("Traffic Type")
plt.ylabel("Number of Records")
plt.show()

#NumPy analysis
duration_array = np.array(df["duration_ms"])
print("\nDuration Values:",duration_array)
print("Mean Duration:",np.mean(duration_array))
print("Maximum Duration:",np.max(duration_array))
print("Minimum Duration:",np.min(duration_array))
```

```
#Select features and target
X =df.drop(["connection_id","label"],axis=1)
y =df["label"]

#Split data
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)

#Train model
model=DecisionTreeClassifier()
model.fit(X_train,y_train)

#Predict
y_pred=model.predict(X_test)

print("\nPredicted Values:")
print(y_pred)

print("\nActual Values:")
print(y_test.values)

#Accuracy
accuracy=accuracy_score(y_test,y_pred)
print("\nAccuracy:",accuracy)
```

```
First 5 rows:
   Unnamed: 0  connection_id  duration_ms  src_bytes  dst_bytes  \
0           0              1          120         10          8
1           1              2          350         45         30
2           2              3          800        150        120
3           3              4           60          5          4
4           4              5         1200        300        260

   failed_logins  packet_count  error_count  suspicious_score  is_encrypted  \
0              0             2            0              0.02             1
1              1             5            2              0.15             1
2              4            20           15              0.85             0
3              0             1            0              0.01             1
4              8            35           22              0.92             0

    label
0  normal
1  normal
2  attack
3  normal
4  attack

Datasetsize:
(5, 11)

Column names:
Index(['Unnamed: 0', 'connection_id', 'duration_ms', 'src_bytes', 'dst_bytes',
       'failed_logins', 'packet_count', 'error_count', 'suspicious_score',
       'is_encrypted', 'label'],
      dtype='str')

Dataset information:
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 11 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Unnamed: 0        5 non-null      int64
 1   connection_id     5 non-null      int64
 2   duration_ms       5 non-null      int64
 3   src_bytes         5 non-null      int64
 4   dst_bytes         5 non-null      int64
 5   failed_logins     5 non-null      int64
 6   packet_count      5 non-null      int64
 7   error_count       5 non-null      int64
 8   suspicious_score  5 non-null      float64
 9   is_encrypted      5 non-null      int64
 10  label             5 non-null      str
dtypes: float64(1), int64(9), str(1)
memory usage: 602.0 bytes
None

Statistical summary:
       Unnamed: 0  connection_id  duration_ms   src_bytes  dst_bytes  \
count    5.000000       5.000000     5.000000    5.000000    5.00000
mean     2.000000       3.000000   506.000000  102.000000   84.40000
```

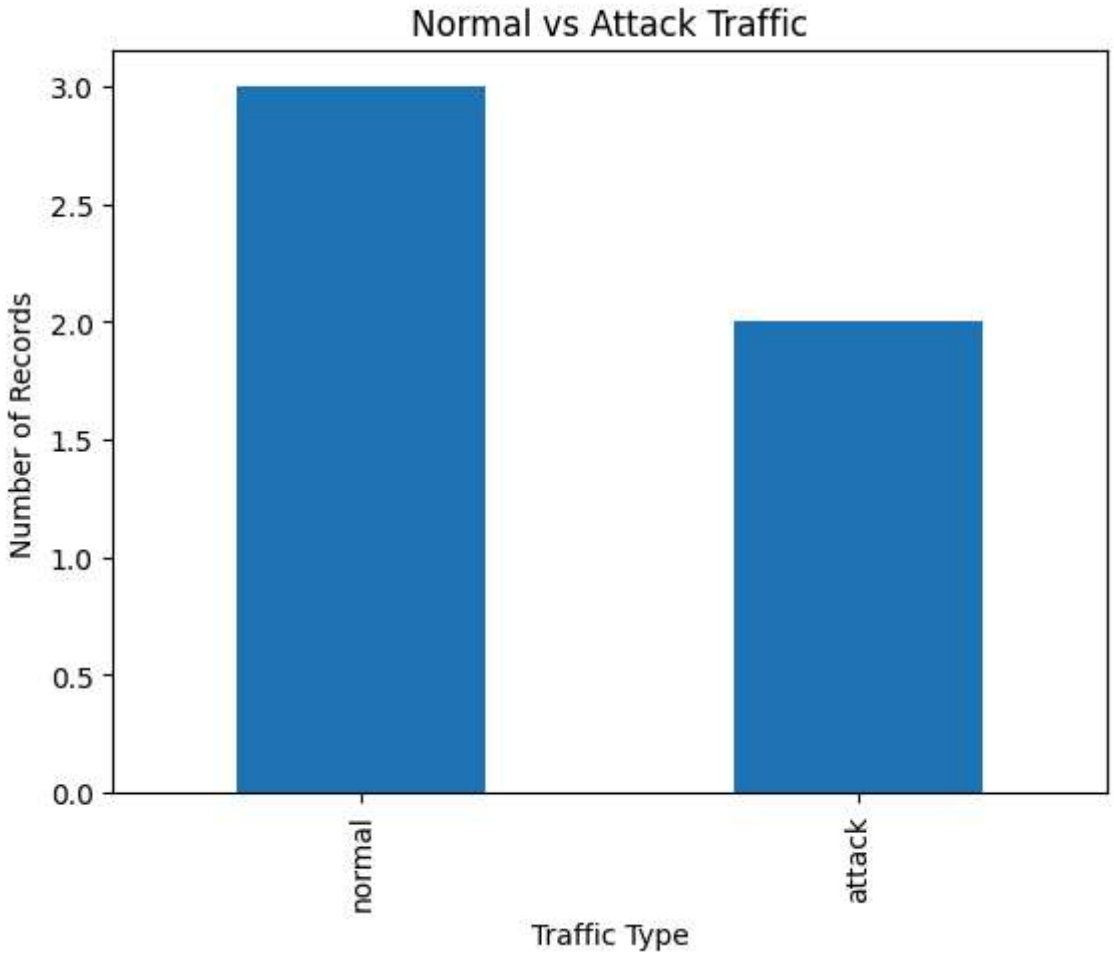
```
std      1.581139      1.581139   484.850492  125.129932  108.81544
min      0.000000      1.000000    60.000000    5.000000    4.00000
25%      1.000000      2.000000   120.000000   10.000000    8.00000
50%      2.000000      3.000000   350.000000   45.000000   30.00000
75%      3.000000      4.000000   800.000000  150.000000  120.00000
max      4.000000      5.000000  1200.000000  300.000000  260.00000

       failed_logins  packet_count  error_count  suspicious_score  \
count       5.000000      5.000000     5.000000          5.000000
mean        2.600000     12.600000     7.800000          0.390000
std         3.435113     14.673105    10.109402          0.455906
min         0.000000      1.000000     0.000000          0.010000
25%         0.000000      2.000000     0.000000          0.020000
50%         1.000000      5.000000     2.000000          0.150000
75%         4.000000     20.000000    15.000000          0.850000
max         8.000000     35.000000    22.000000          0.920000

       is_encrypted
count      5.000000
mean       0.600000
std        0.547723
min        0.000000
25%        0.000000
50%        1.000000
75%        1.000000
max        1.000000

Missing values:
Unnamed: 0          0
connection_id       0
duration_ms         0
src_bytes           0
dst_bytes           0
failed_logins       0
packet_count        0
error_count         0
suspicious_score    0
is_encrypted        0
label               0
dtype: int64

Class distribution:
label
normal    3
attack    2
Name: count, dtype: int64
```




```
Duration Values: [ 120  350  800   60 1200]
Mean Duration: 506.0
Maximum Duration: 1200
Minimum Duration: 60

Predicted Values:
['normal' 'attack']

Actual Values:
<ArrowStringArray>
['normal', 'attack']
Length: 2, dtype: str

Accuracy: 1.0
```

In [ ]: